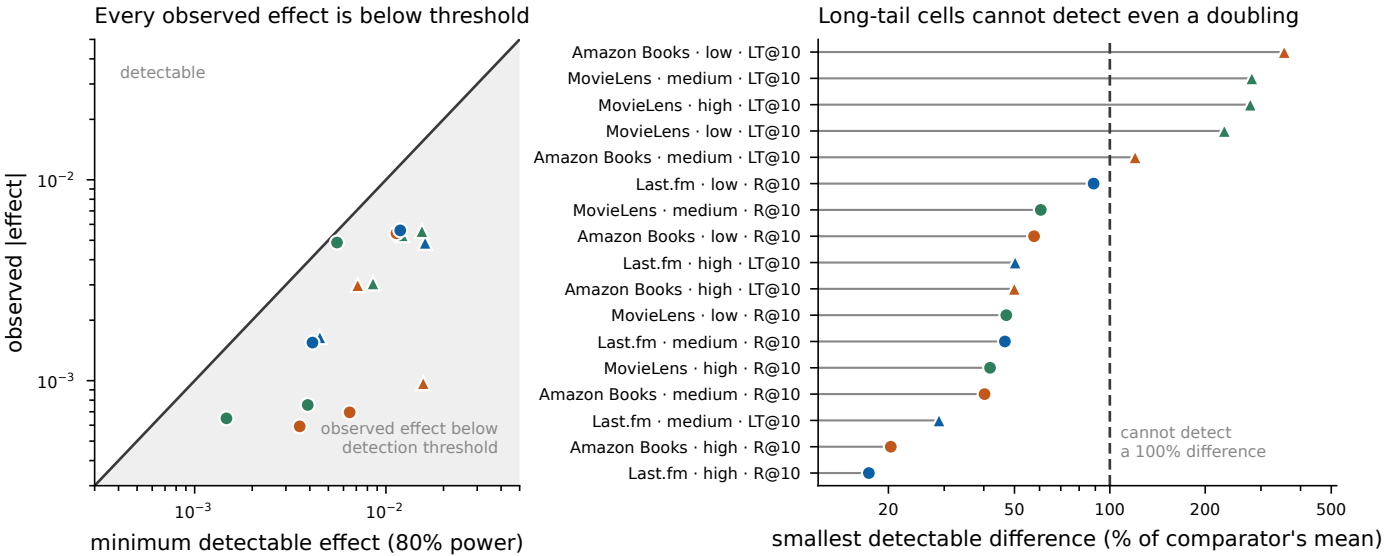
Every observed effect is below threshold
detectable
observed effect below
detection threshold
observed |effect|
10^{-2}
10^{-3}
10^{-3}
10^{-2}
minimum detectable effect (80% power)
Long-tail cells cannot detect even a doubling
Amazon Books · low · LT@10
MovieLens · medium · LT@10
MovieLens · high · LT@10
MovieLens · low · LT@10
Amazon Books · medium · LT@10
Last.fm · low · R@10
MovieLens · medium · R@10
Amazon Books · low · R@10
Last.fm · high · LT@10
Amazon Books · high · LT@10
MovieLens · low · R@10
Last.fm · medium · R@10
MovieLens · high · R@10
Amazon Books · medium · R@10
Last.fm · medium · LT@10
Amazon Books · high · R@10
Last.fm · high · R@10
cannot detect
a 100% difference
20
50
100
200
500
smallest detectable difference (% of comparator's mean)
Last.fm-1K
Amazon Books
MovieLens-25M
Recall@10
LT-Recall@10